

UNIVERSITY INQUIRY AND LOST PROPERTY SYSTEM

FINAL PROJECT REPORT

University Service Management Web Application

ICT 1209 – Web Technologies

Group Members & Registration Numbers/Index Numbers:

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Submitted Date: 2026/09/27

Project Overview

- **Purpose:**

The purpose of this project is to develop a web-based application that helps university students and staff report lost and found items through a centralized online platform. The system also allows users to submit inquiries related to lost property and university services, aiming to improve communication, simplify the reporting process, and make it easier to manage lost property records.

- **Target Audience:**

- University Students
- Academic Staff
- Non-Academic Staff
- University Visitors

- **Key Features:**

- 1) User Registration and Login: Secure account creation and authentication using password hashing (password_hash) and sessions.
- 2) Report Lost & Found Property: Users can submit details (item name, category, location, date, description, and images) of items they lost or found.
- 3) Search & Filter Items: Quick searching and filtering capabilities to find lost and found items easily.
- 4) Online Inquiries / Contact Form: Allows users to submit inquiries with automatic database saving and confirmation messages.
- 5) Dashboard / My Reports: Users can view, manage, edit, or delete their personal lost and found reports.

System Design

- **Database Schema Definition**

The system utilizes a relational MySQL database named `foundone_db` which comprises three primary tables to handle application workflows:

1. **users Table:** Stores registered user accounts including unique user IDs, usernames, email addresses, created timestamps, and securely hashed passwords generated via BCRYPT.
2. **messages Table:** Holds user inquiries submitted through the contact form, including user name, email address, message body, and submission timestamp.
3. **lost_found_items Table:** Stores lost and found item records, containing details such as item title, category, description, location, lost/found status, reported date, and associated image filename.

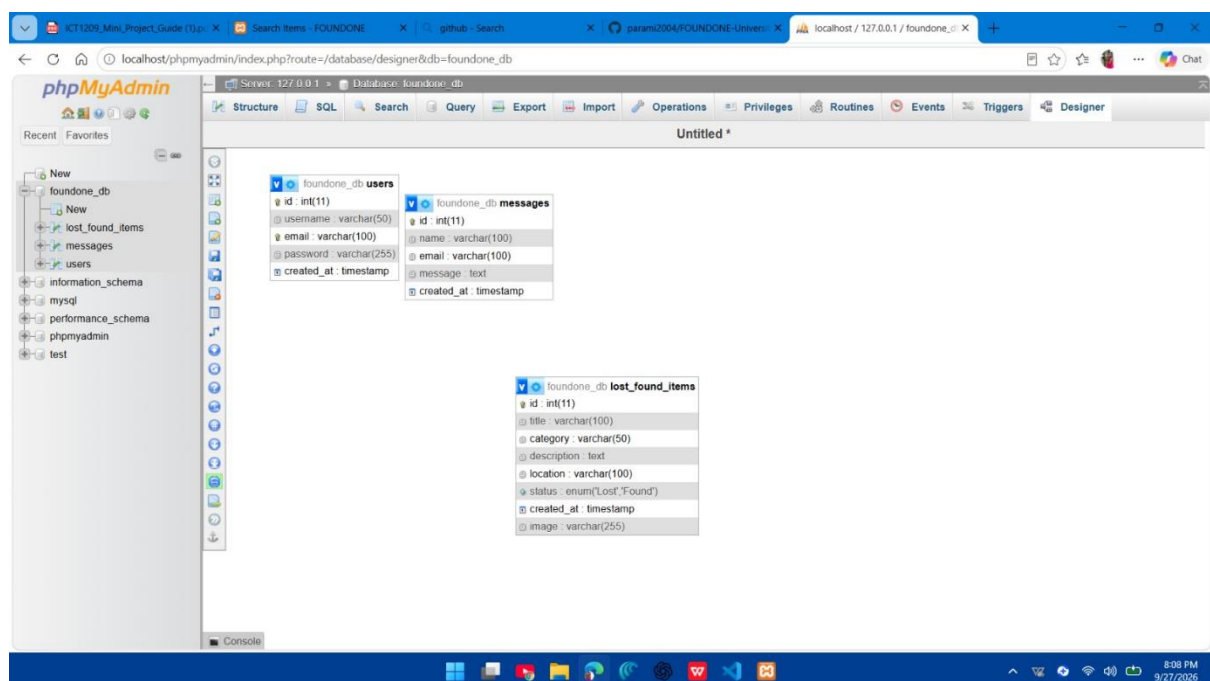


Figure 1: Database Schema / ER Diagram

- **Directory & Folder Architecture**

The project structure strictly adheres to modular coding guidelines to ensure maintainability and security:

1. **auth:** Contains authentication controllers (register.php, login.php, logout.php).
2. **includes:** Houses shared utility scripts such as PDO database configuration (db.php) and helper functions (functions.php).
3. **css & /js:** Contains custom style definitions (style.css) and client-side form validation scripts (main.js)
4. **uploads:** Dedicated directory for storing uploaded item images.
5. **Root Files:** Core pages including index.php, report.php, search.php, dashboard.php, and contact.php.

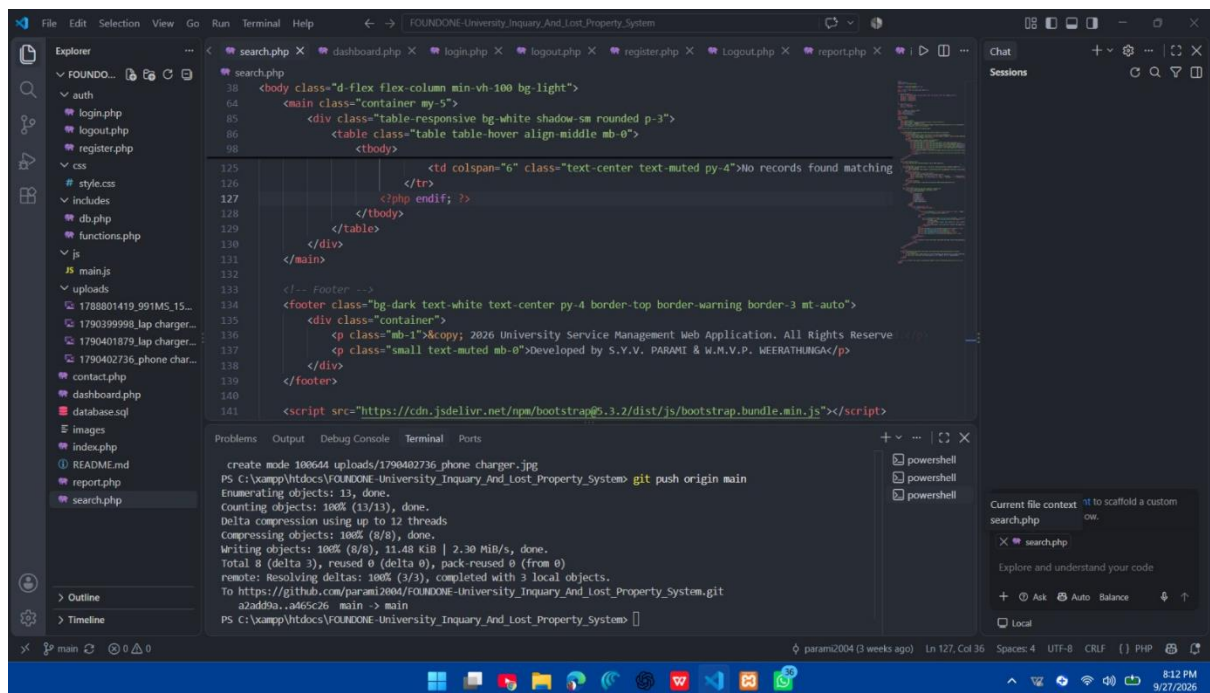


Figure 2: Project Folder Structure

Implementation

User Authentication & Password Hashing

The application provides secure user onboarding and authentication through dedicated login and registration modules:

- **User Registration (auth/register.php):** Captures dynamic user credentials. To maintain data security, passwords are strictly hashed using PHP's native password_hash() function with the PASSWORD_BCRYPT algorithm before being persisted to the MySQL database.
- **User Login & Session Management (auth/login.php):** Authenticates submitted credentials against hashed values using password_verify(). Upon successful authentication, a secure session is initialized, and session_regenerate_id() is invoked to prevent session fixation attacks.
- **Logout (auth/logout.php):** Clears session data and destroys active user sessions cleanly upon sign-out.

Home Page & Navigation:

Built using Bootstrap 5 navbar, hero section, and search layout.

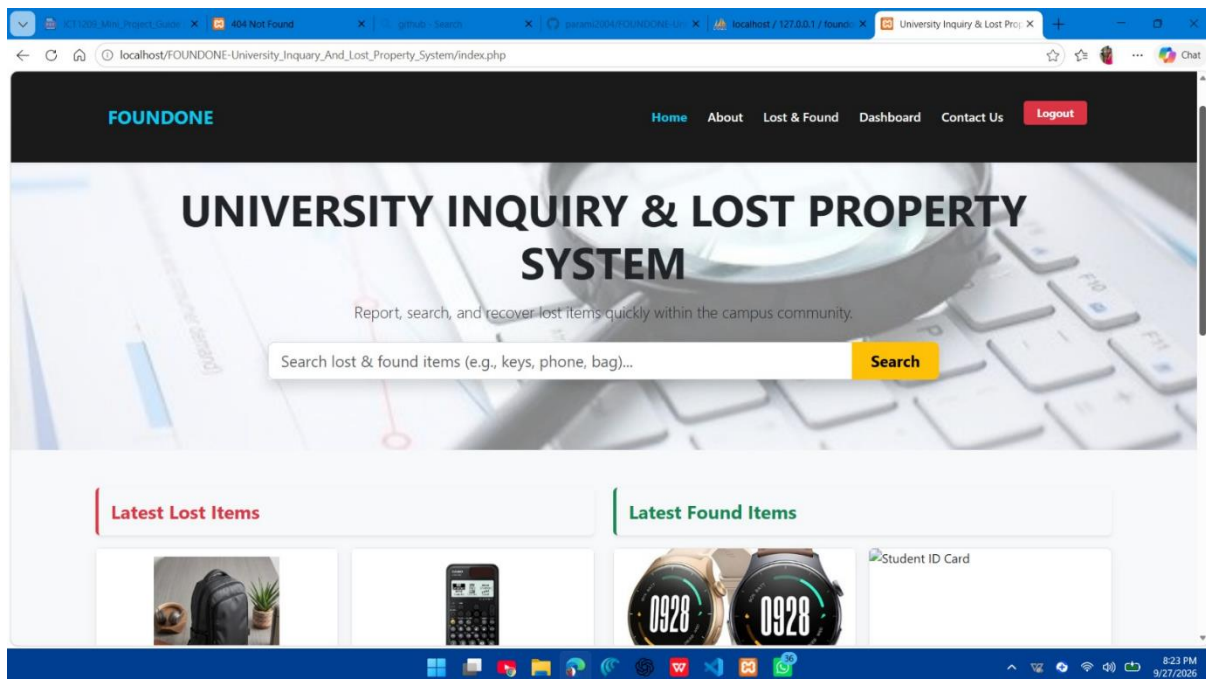


Figure 3: Home Page (index.php):

Lost/Found Item Form (report.php):

Forms capture item details and insert records into MySQL linked to the logged-in user.

Report Lost & Found Item

Fill out the form below to report a lost or found item within the campus community.

Item Name *
e.g., Scientific Calculator, Black Laptop Bag

Report Type *
Choose... Lost or Found?

Category *
Select Category

Date *
09/27/2026

Location *
e.g., Main Library, IT Faculty Room 2

Description *
Provide details like color, brand, unique marks...

Upload Image (Optional)
Choose File No file chosen

Figure 4: Report Lost/Found Item Form

Showing rows 0 - 4 (5 total, Query took 0.0013 seconds)

SELECT * FROM `lost\_found\_items`

Query results operations: Print, Copy to clipboard, Export, Display chart, Create view

id	title	category	description	location	status	created_at	image
1	lap charger	Electronics	black colour MSI charger	library	Lost	2026-09-06 12:09:30	NULL
2	lap bag	Clothing	gray colour bag	S502 hall, Genesis building	Found	2026-09-06 22:26:38	NULL
3	scientific Calculator	Electronics	gray colour calculator with red colour sticker	faculty canteen	Lost	2026-09-07 22:46:59	NULL
5	laptop charger	Electronics	black color MSI charger	Main IT lab ,3rd floor	Lost	2026-09-26 11:21:19	NULL
6	phone charger	Electronics	white colour c pin charger	library	Found	2026-09-26 11:35:36	NULL

Figure 5: Lost/Found Item Form (report.php)

Search / Item List Page (search.php):

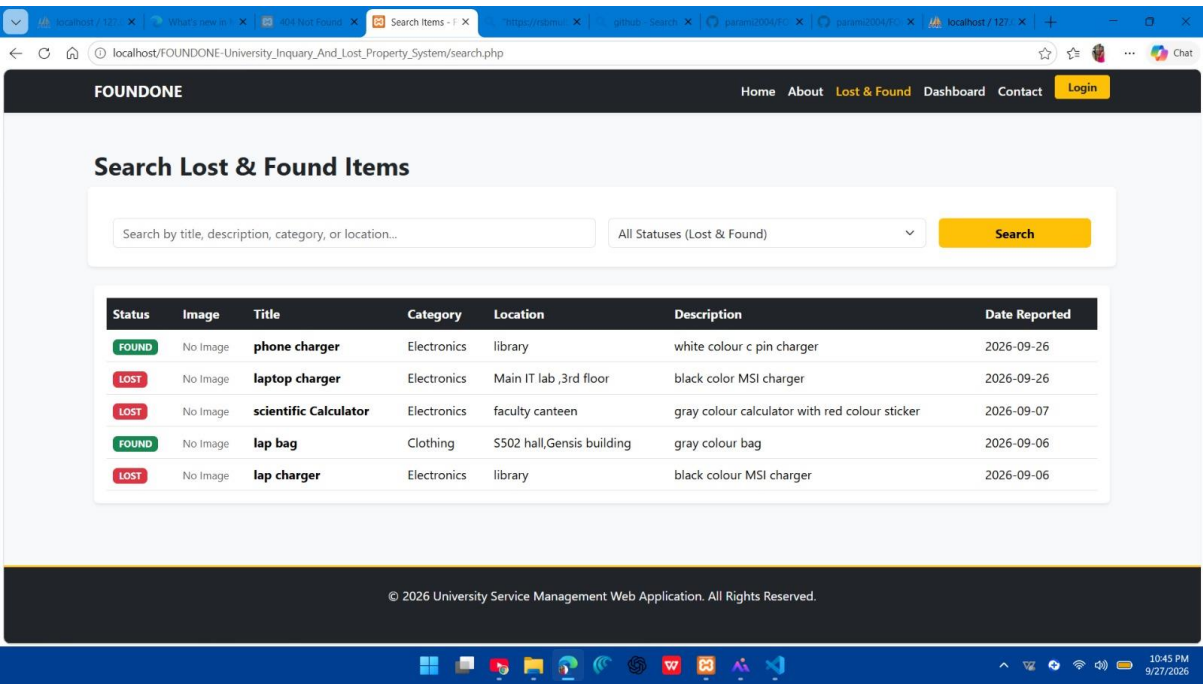


Figure 6:Search / Item List Page

Dashboard Page (dashboard.php):

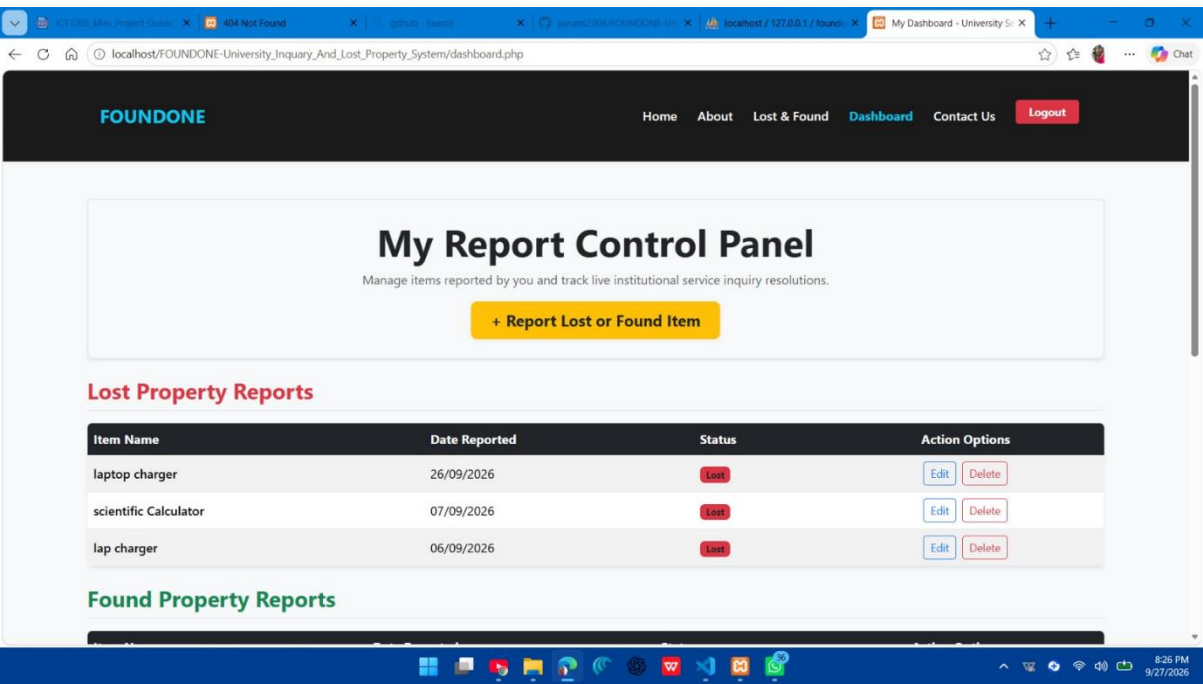


Figure 7:Dashboard Page

Contact Page (contact.php):

Processes inquiries via contact.php, stores them in the messages table, and displays a success alert.

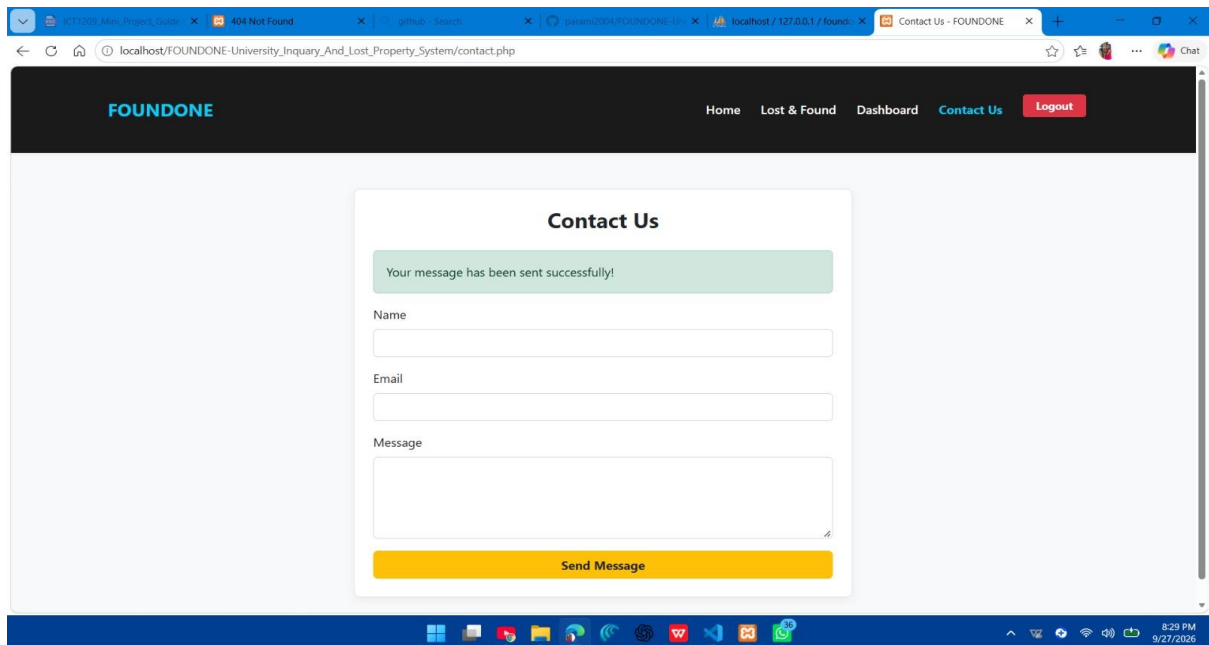


Figure 8: Contact Page

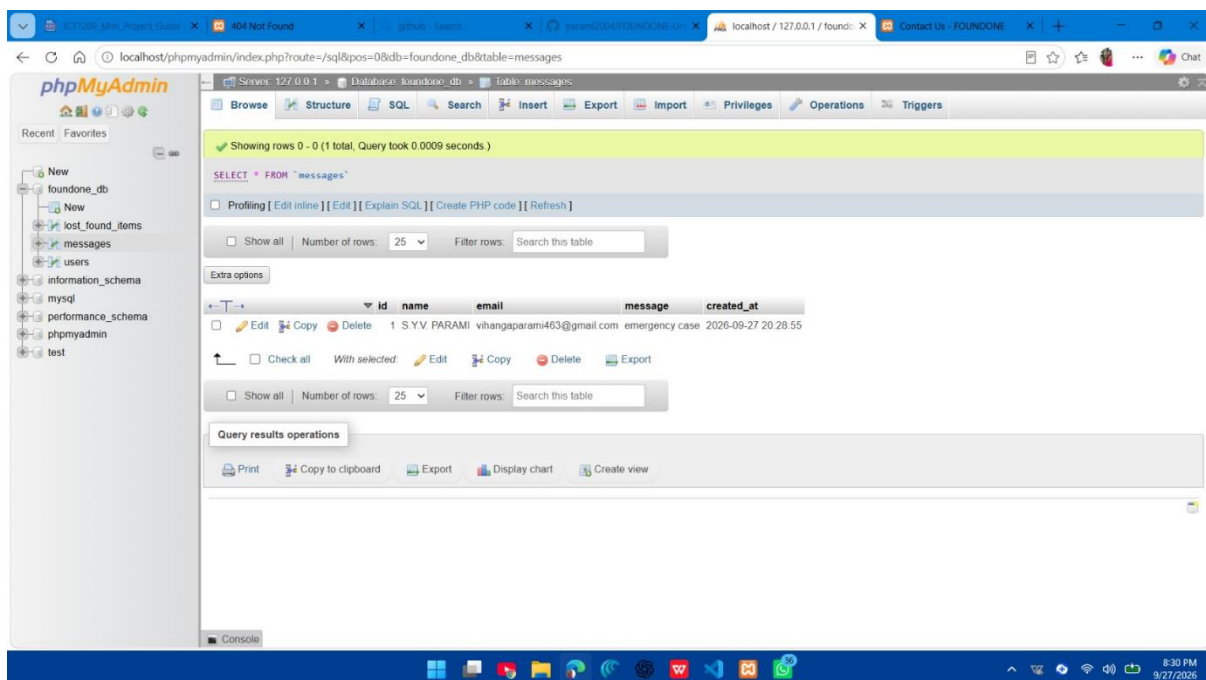


Figure 9: Contact Page (contact.php)

Report lost and found / Mobile View:

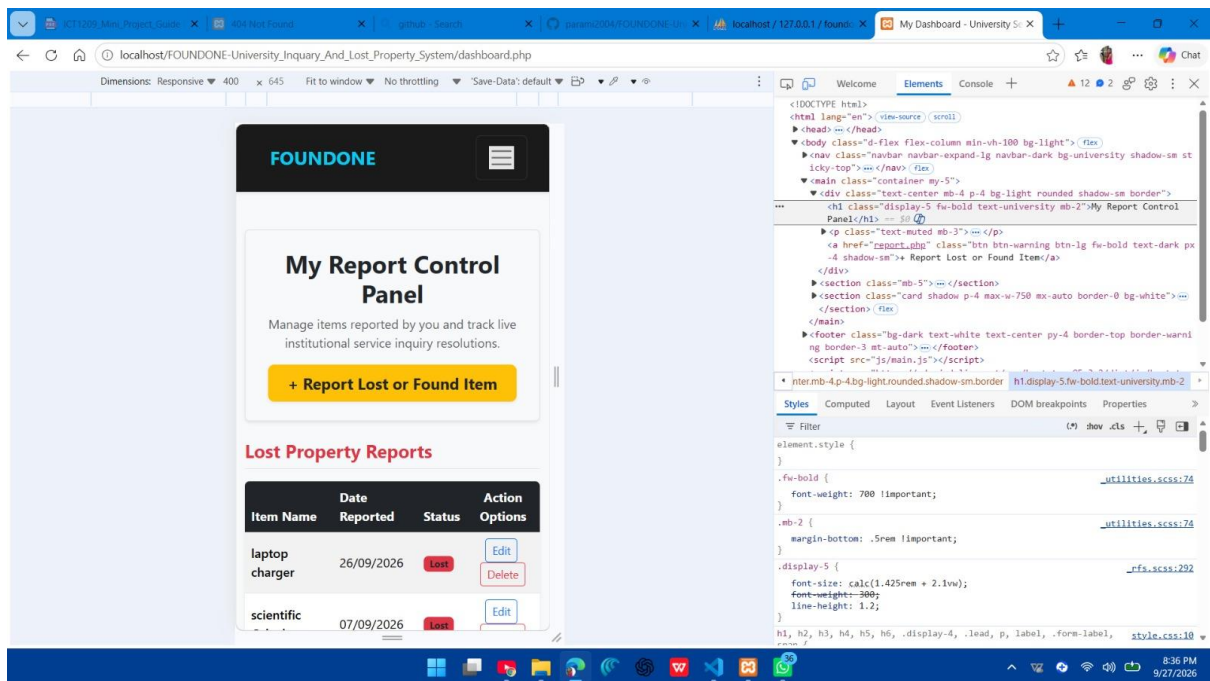


Figure 10: Report lost and found / Mobile View

Database Schema / ER Diagram

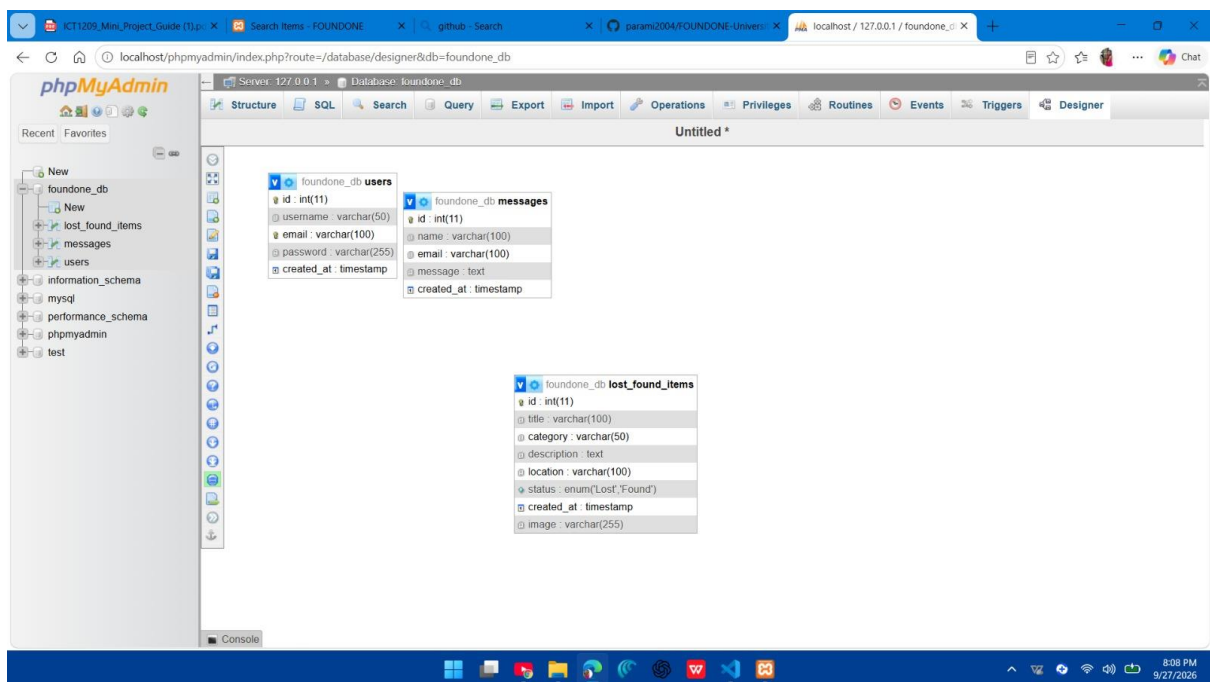


Figure 11: Database Schema / ER Diagram

Challenges and Solutions

- **Challenge 1: Preventing SQL Injections:** Direct user inputs in database queries posed security vulnerabilities.
 - ✓ **Solution:** Used PHP Prepared Statements (PDO / MySQLi) for all database operations.
- **Challenge 2: Secure Session Management:** Handling user login states securely without unauthorized access.
 - ✓ **Solution:** Implemented `password_hash()` for passwords and `session_regenerate_id()` upon successful login, combined with clean session destruction during logout.
- **Challenge 3: Responsive UI Designing:** Making data tables and form grids adapt correctly to mobile and tablet screens.
 - ✓ **Solution:** Utilized Bootstrap 5 responsive grid classes (container, row, col-md-*) and flexible CSS utility classes.

Individual Contribution

1. S.Y.V. PARAMI [ITT/2024/077]:

- Designed the overall database schema and configured the XAMPP MySQL environment.
- Developed Backend PHP scripts for User Authentication (Registration, Login, Logout) and secure Session Management.
- Implemented client-side JavaScript features including form validations, dynamic content toggling, and search filtering.

2. W.M.V.P. WEERATHUNGA [ITT/2024/116]:

- Designed and developed the Frontend layout (covering Home, Login, Register, Dashboard, and Contact pages).
- Compiled the project documentation.
- Implemented the Contact/Inquiry form backend processing and database integration.

Setup Instructions

To run this project locally on your machine, follow these steps:

1. **Prerequisites:** Install **XAMPP** (or WAMP) with Apache and MySQL enabled.

2. **Database Setup:**

- Start Apache and MySQL from the XAMPP Control Panel.
- Open your browser and go to <http://localhost/phpmyadmin/>.
- Create a new database named `university_lost_property_db`.
- Import the provided `database.sql` file into this database.

3. **Project Deployment:**

Copy the entire project folder into the `htdocs` directory of your XAMPP installation (e.g., `C:\xampp\htdocs\project`).

4. **Running the Application:**

<http://localhost/project/index.php>

References

- Bootstrap 5 Official Documentation – <https://getbootstrap.com/>
- PHP 8 Manual and Functions Guide – <https://www.php.net/>
- W3Schools Web Development Tutorials – <https://www.w3schools.com/>
- MySQL Database Documentation – <https://dev.mysql.com/doc/>